

DIN-SQL: Original Prompting Strategy vs. Dynamic Demonstration Retrieval

A comparison of fixed vs. semantically-retrieved few-shot examples

This document explains the key limitation in the original DIN-SQL paper's prompting approach and shows — with concrete examples — how Dynamic Demonstration Retrieval addresses it.

1. What the Original Paper Does

DIN-SQL decomposes text-to-SQL into four LLM calls: Schema Linking, Classification, SQL Generation, and Self-Correction. Each call is a *few-shot prompt* — the LLM is shown several examples before being asked to handle the real query. The critical design choice is **how those examples are chosen**.

The answer from the paper: they are hand-picked, fixed forever.

The authors manually selected a small set of question/SQL pairs from the Spider training set and **hardcoded** them into each prompt as static strings in the source code. These same examples are injected into every single prompt, for every single test query — regardless of domain, complexity, or vocabulary.

Module	Fixed examples	Source databases
Schema Linking	10 (always the same)	farm_competition, dept_mgmt, student_transcripts, soccer_1, university
Classification	10 (always the same)	university only
SQL Gen — Easy	14 (always the same)	university only
SQL Gen — Non-Nested	7 (always the same)	university only
SQL Gen — Nested	10 (always the same)	university only
Self-Correction	0 (zero-shot)	—

Key stat: only ~5 out of 146 training databases were ever used as demonstration sources. The other 141 databases — and their SQL patterns — were completely ignored during prompt construction.

2. A Concrete Example of the Problem

Suppose the test query is about a **hospital database**:

Test Question

"Find all patients admitted after January 2020 who have a diagnosis of diabetes and were treated by a cardiologist."

What the original DIN-SQL injects into the prompt:

The LLM receives these fixed university examples — completely unrelated to hospitals:

Q: "Find the buildings which have rooms with capacity more than 50."

SQL: SELECT DISTINCT building FROM classroom WHERE capacity > 50

Q: "What is the name of the department with the most credits?"

SQL: SELECT dept_name FROM course GROUP BY dept_name ORDER BY sum(credits) DESC LIMIT 1

Q: "How many instructors teach a course in the Spring of 2010?"

SQL: SELECT COUNT(DISTINCT ID) FROM teaches WHERE semester = 'Spring' AND YEAR = 2010

Q: "Find the name of students who took any class in 2009 and 2010."

SQL: SELECT DISTINCT T1.name FROM student AS T1 JOIN takes AS T2 ON T1.id = T2.id WHERE T2.YEAR = 2009 OR T2.YEAR = 2010

The problem: The LLM sees examples about classrooms, departments, and course credits — none of which share vocabulary, table structure, or query patterns with the hospital question. The model must bridge a large semantic gap entirely on its own.

What Dynamic Retrieval injects instead:

FAISS finds training examples semantically closest to the hospital query:

Q: "List all patients who were admitted after 2018."

SQL: SELECT * FROM patients WHERE admission_date > '2018-01-01'

Q: "Find employees who have the role of cardiologist in department 3."

SQL: SELECT name FROM employees WHERE role = 'cardiologist' AND dept_id = 3

Q: "How many records have a diagnosis matching type 2 diabetes?"

SQL: SELECT COUNT(*) FROM records WHERE diagnosis = 'type 2 diabetes'

Q: "Find patients treated by a specialist after a given date."

SQL: SELECT T1.name FROM patients T1 JOIN treatments T2 ON T1.id = T2.patient_id WHERE T2.date > '2020-01-01'

The benefit: Every example shares date-filtering, role-matching, and JOIN patterns with the test query. The LLM receives directly transferable structural guidance.

3. Side-by-Side: Three More Contrasting Examples

Concert & Music Domain

Test Question	""How many singers are from France?""	
	<i>Original</i> (fixed, unrelated)	<i>Dynamic Retrieval</i> (semantically matched)
	Q: "How many instructors teach in Spring 2010?" ← university, unrelated Q: "Find buildings with rooms over 50 capacity." ← university, unrelated	Q: "How many artists released an album in France?" ← music, closely related Q: "Count the number of performers by country." ← same domain pattern

E-commerce / Orders Domain

Test Question	""Find the total revenue from orders placed in December 2023.""	
	<i>Original</i> (fixed, unrelated)	<i>Dynamic Retrieval</i> (semantically matched)
	Q: "Find the total budgets of Marketing or Finance dept." ← dept budgets Q: "What is the average salary above 42000?" ← salaries	Q: "What is the total sales amount for November orders?" ← same pattern Q: "Find total revenue grouped by product category." ← same agg type

Multi-table Nested Query

Test Question	""Find students who passed all exams but never attended a lab session.""	
	<i>Original</i> (fixed, unrelated)	<i>Dynamic Retrieval</i> (semantically matched)
	Q: "Find instructors who taught Fall 2009 not Spring 2010." ← unrelated domain Q: "Find courses that do not have any prerequisite." ← different NOT IN pattern	Q: "Find employees who completed training but never took leave." ← same NOT IN + JOIN logic Q: "List members who attended all events but missed the finale." ← matching nested structure

4. Why This Matters: The Paper's Own Words

The authors themselves acknowledged this limitation in Section 7 (Limitations):

*"Our manually constructed demonstrations are fixed for each query class. Future research may explore **adaptive and automated methods** for generating demonstrations at finer granularities, which can further enhance the performance of our approach."*

Dynamic Demonstration Retrieval is the direct implementation of exactly what the authors flagged as future work. The extension is well-motivated, grounded in the paper itself, and addresses a measurable weakness.

5. Summary Comparison

Dimension	Original DIN-SQL	Extended DIN-SQL
Example selection	Manual, hand-picked	Automated via FAISS similarity
Examples per prompt	Fixed 10 (identical every time)	Dynamic top-k (different per query)
Training DBs used	~5 out of 146	All 146 training databases
Domain relevance	Random / often unrelated	Semantically matched to test query
Prompt construction	Static hardcoded string	Built at runtime
Coverage of patterns	University-domain heavy	Matches whatever domain the query is in
Motivation	Baseline	Directly from paper's stated future work